

INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT.

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Admission Number: BIT/2024/74441.

School/faculty: Computing and informatics.

Programme: Degree

Academic Year: III

Semester: II

Lecturer: Michael Nyoro.

Phone Number: 0722313169

WEEK 1.

Tools used:

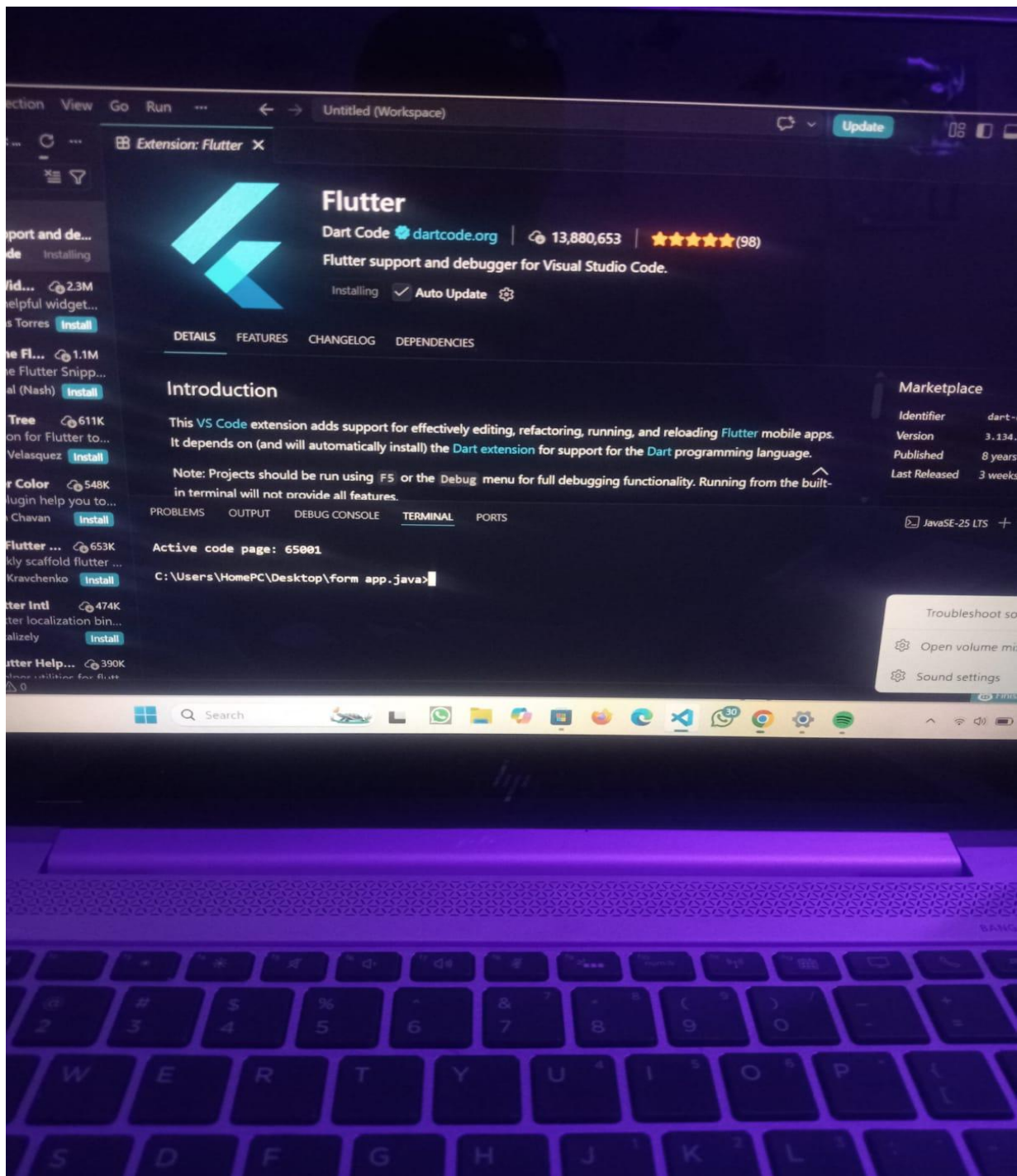
Android studio.

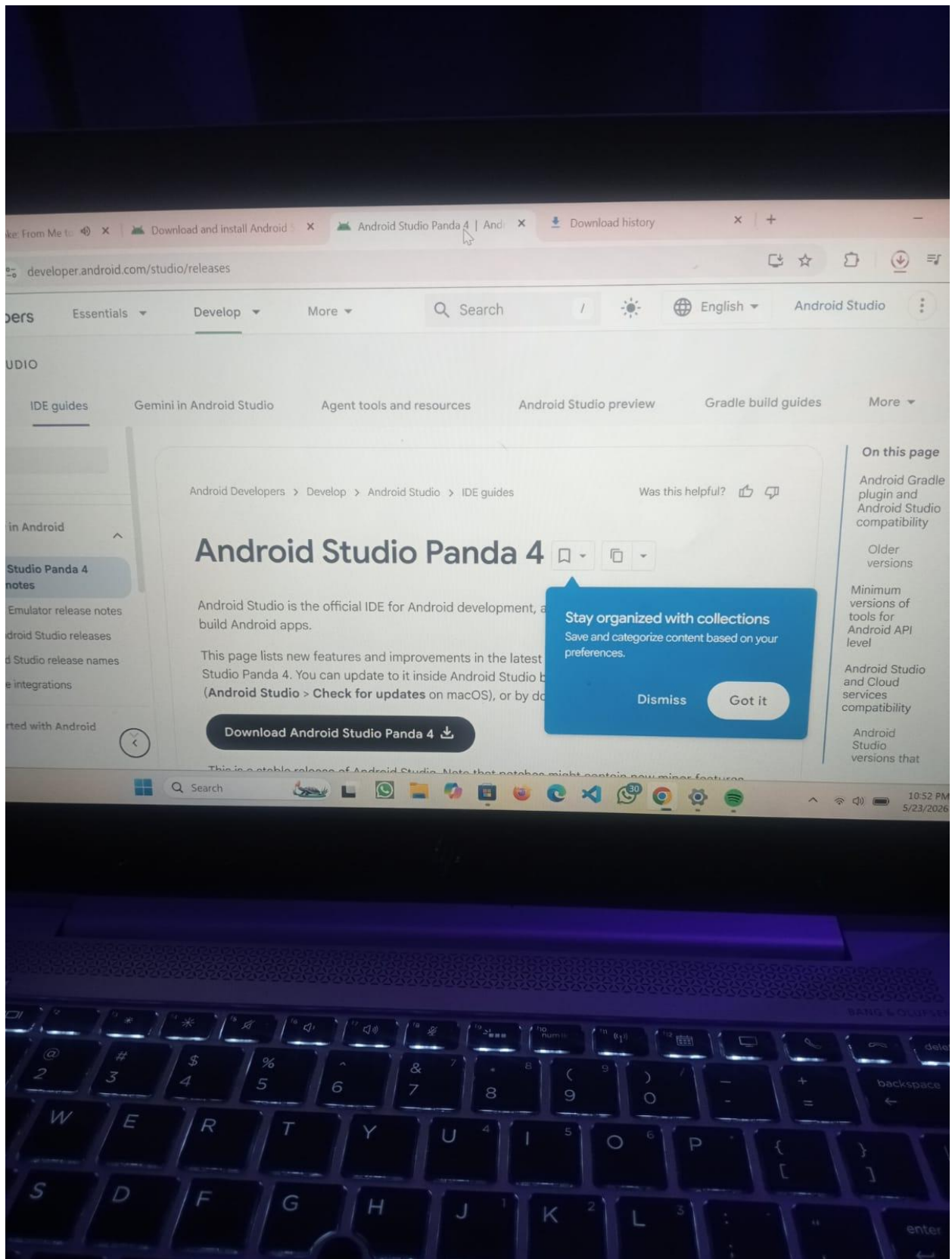
Figma

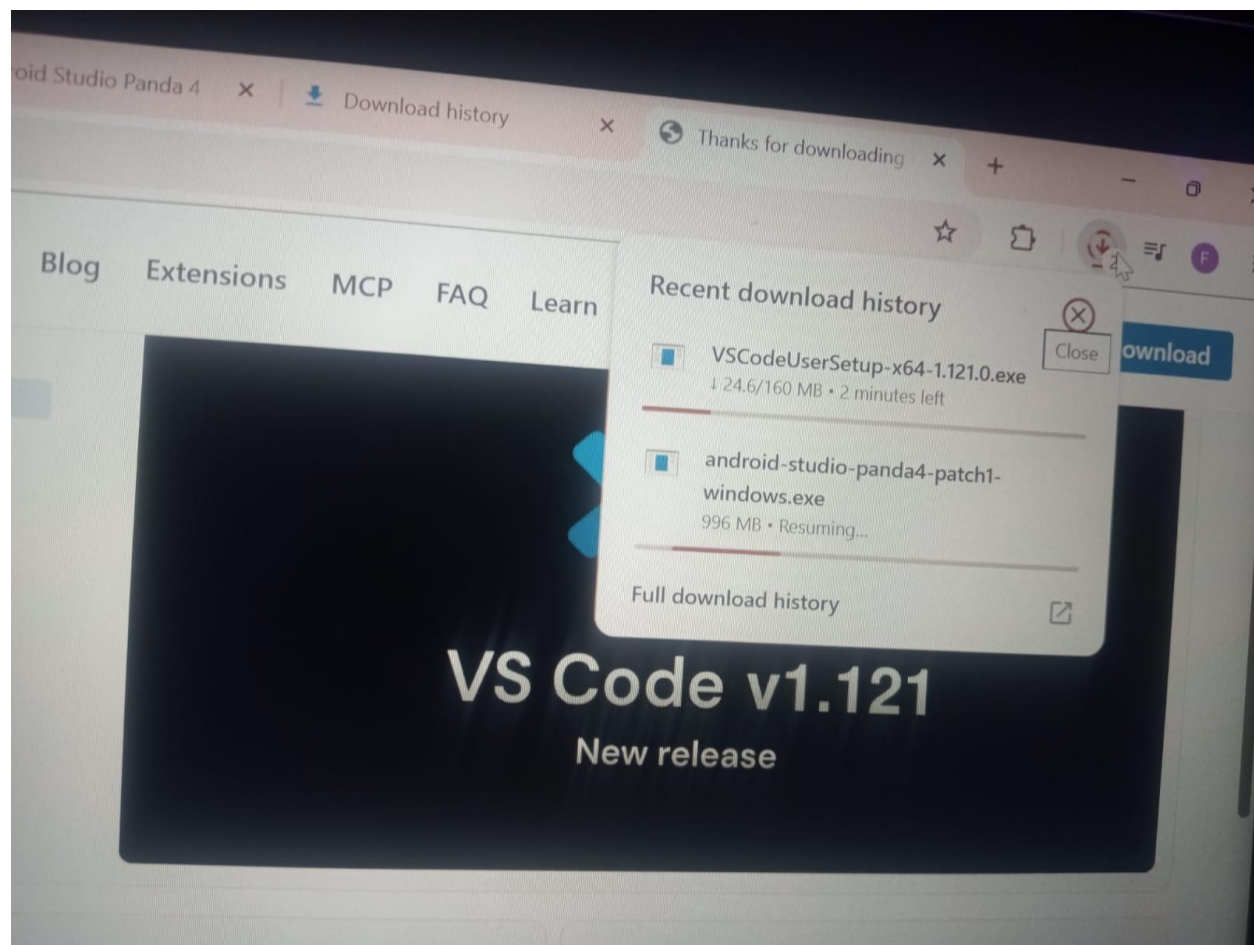
Vs code

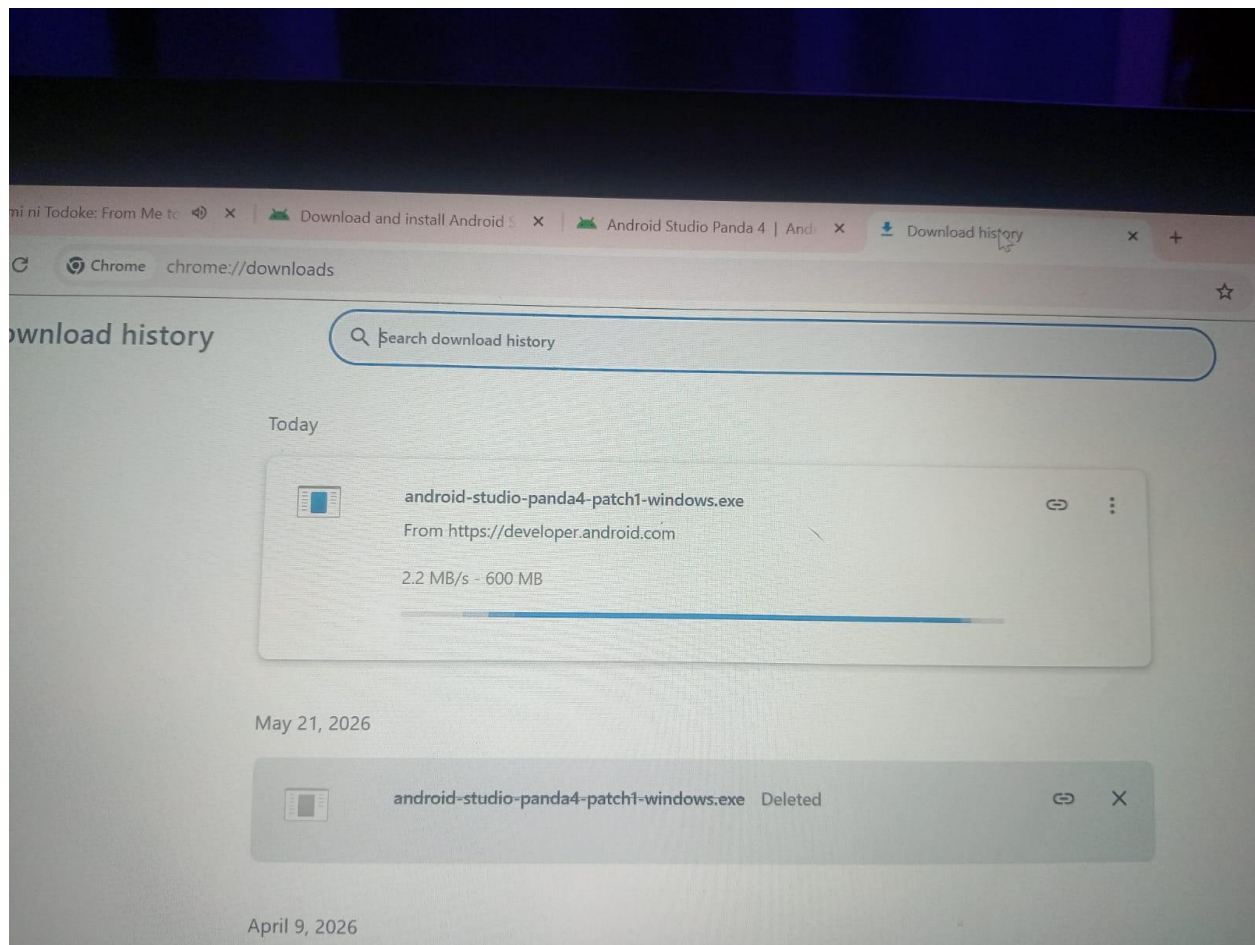
Flutter SDK

Android phone





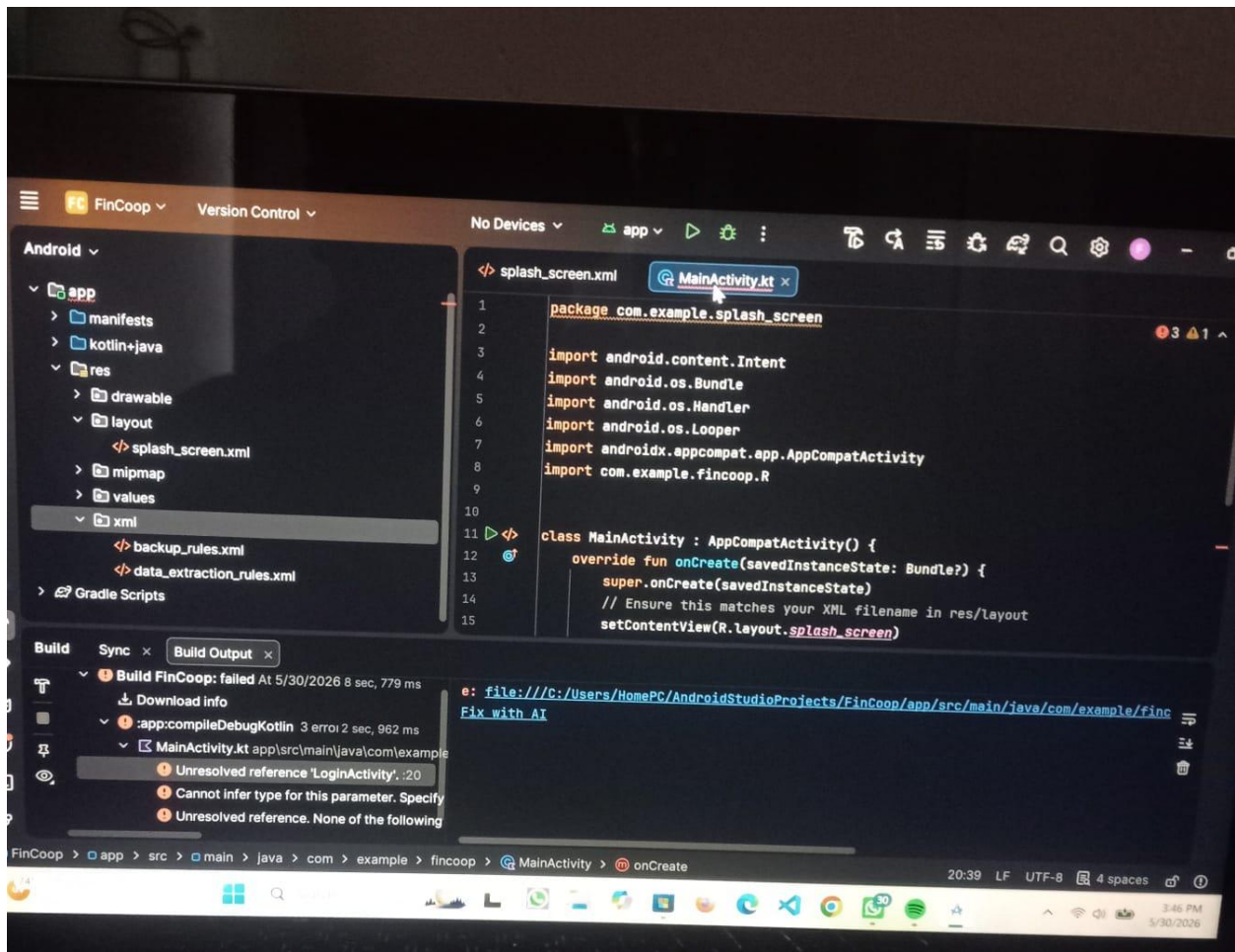


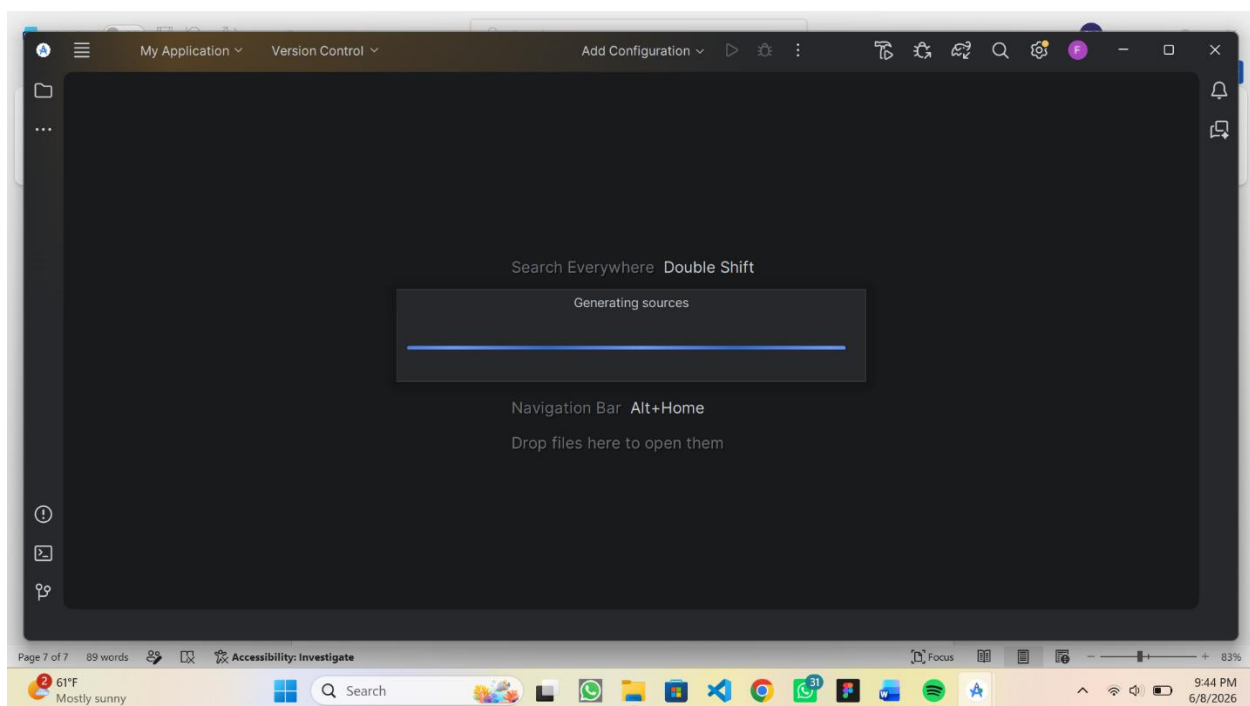
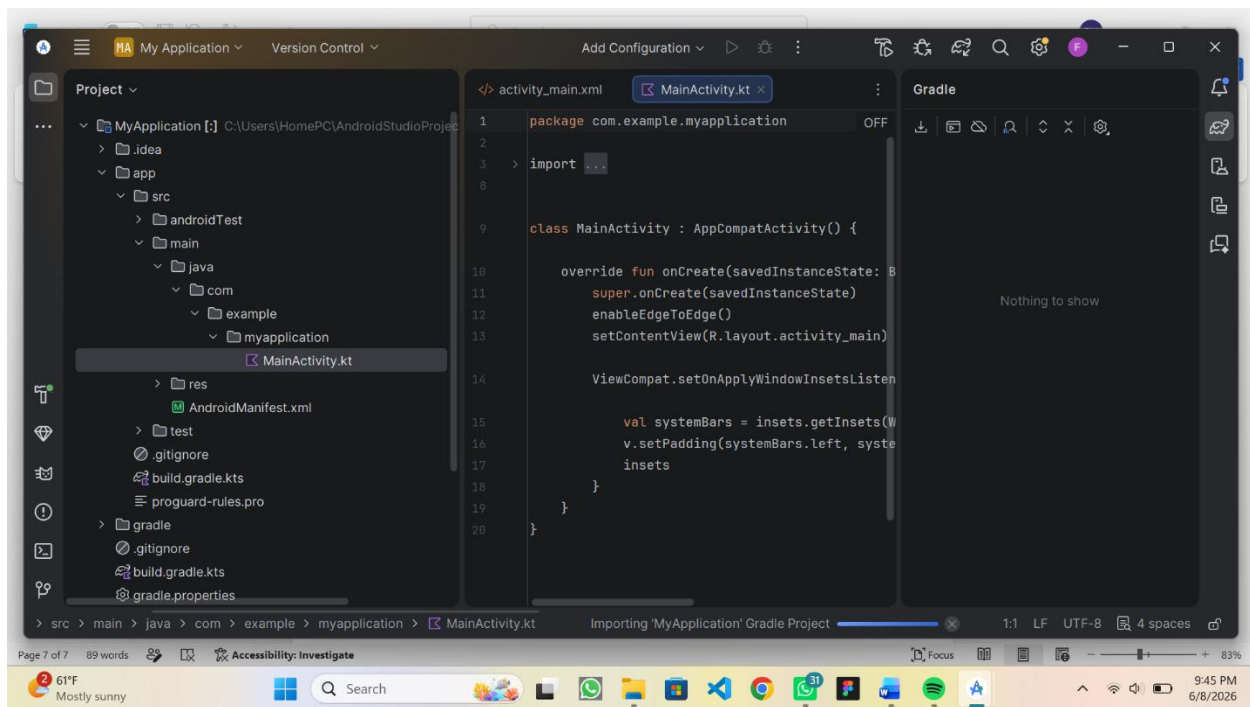


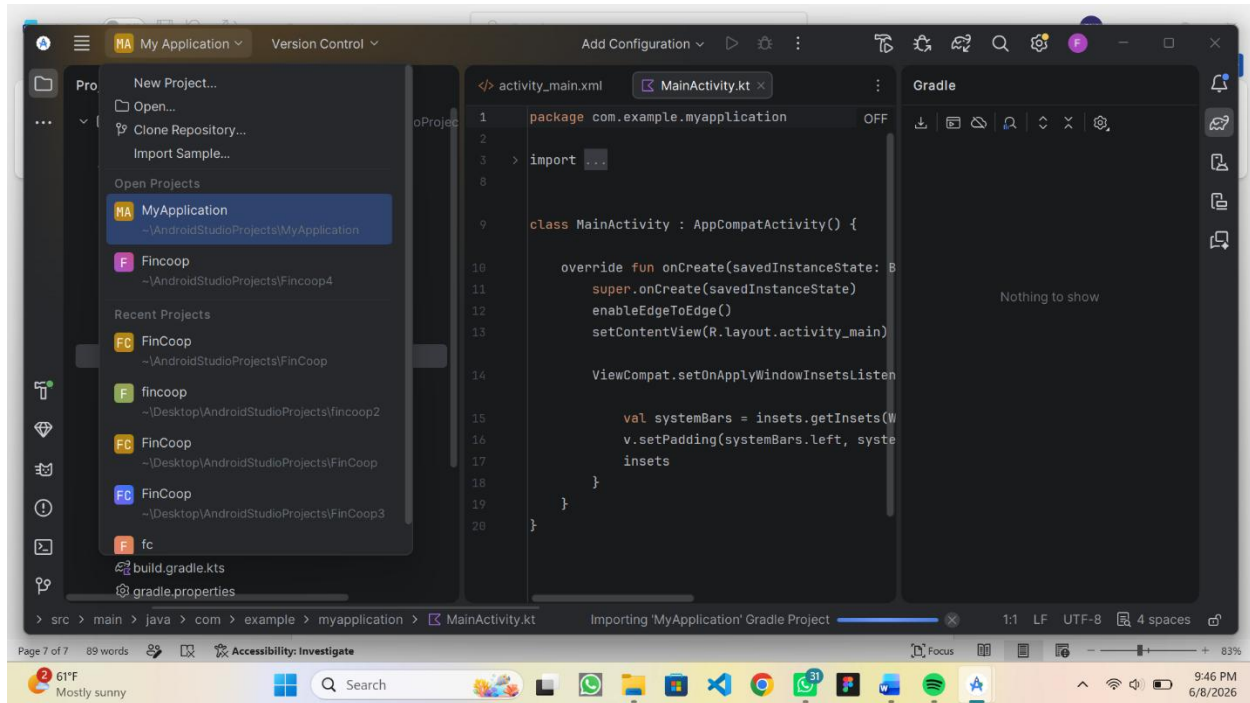
PERSONAL REFLECTION

Installing the tools was challenging at first due to errors and configuration issues, but it improved my problem-solving skills. I gained practical experience in setting up development environments and became more confident in using modern software tools for app development.

WEEK 2: MOBILE DEVELOPMENT ENVIRONMENT.



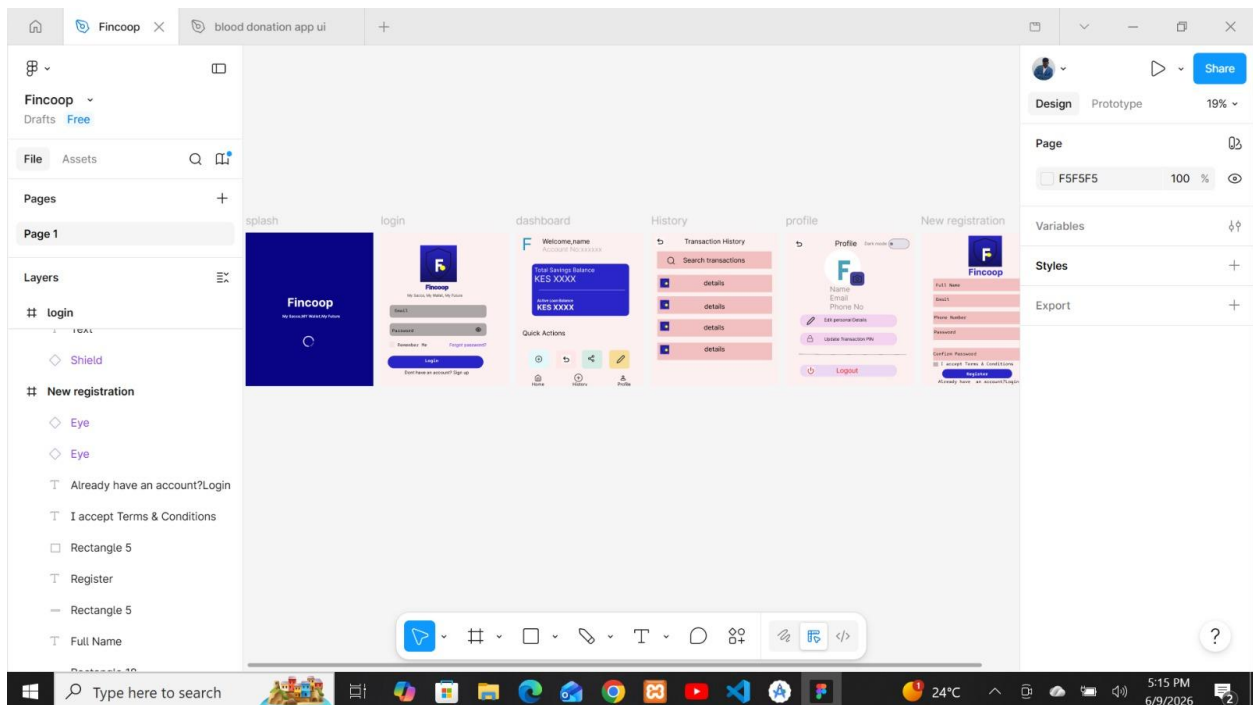
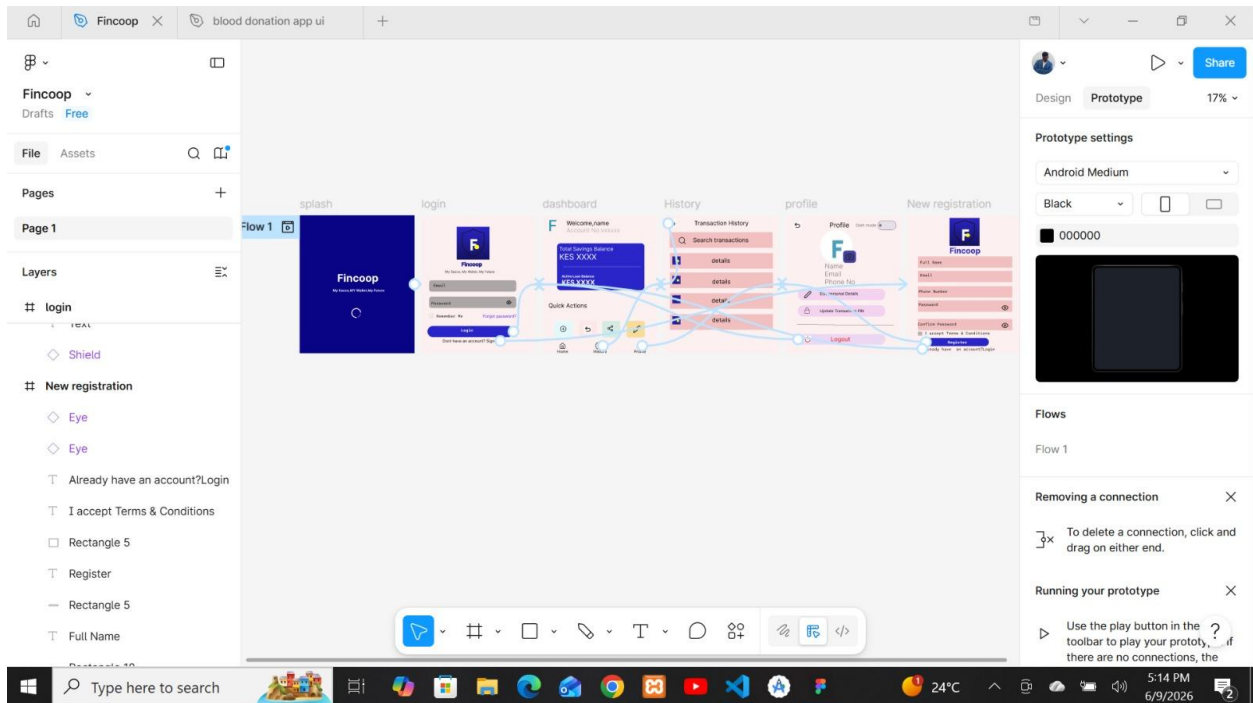


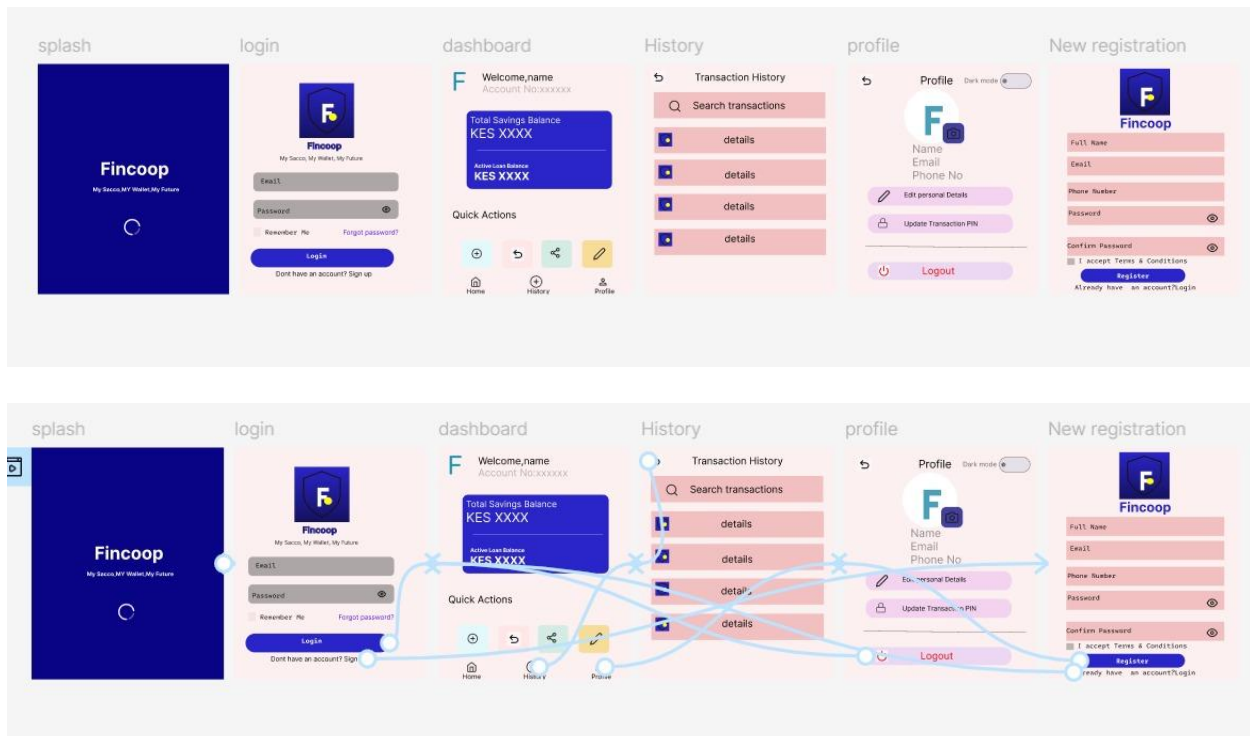


PERSONAL REFLECTION

I set up the mobile development environment by installing and configuring tools like Android Studio, Flutter SDK, and device emulators. This process helped me understand how different components work together, though I faced minor setup challenges that enhanced my troubleshooting and technical skills.

WEEK 3: USER INTERFACE DESIGN





Personal reflection

I focused on user interface design for the fincoop app using Figma. I learned how to create and organize screens, design user-friendly layouts, and improve overall system usability. This improved my understanding of UI/UX principles and enhanced my design skills.

WEEK 4: EVENT HANDLING AND USER INTERACTION.



Imani

imani@gmail.com
+254 700 123 456

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Logout



Fincoop

Full Name

Email

Phone Number

Password



Confirm Password



☐ I accept Terms & Conditions

Register

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Dark Mode



Fincoop

My Sacco, My Wallet, My Future

Email

Password



☐ Remember Me

[Forgot Password?](#)

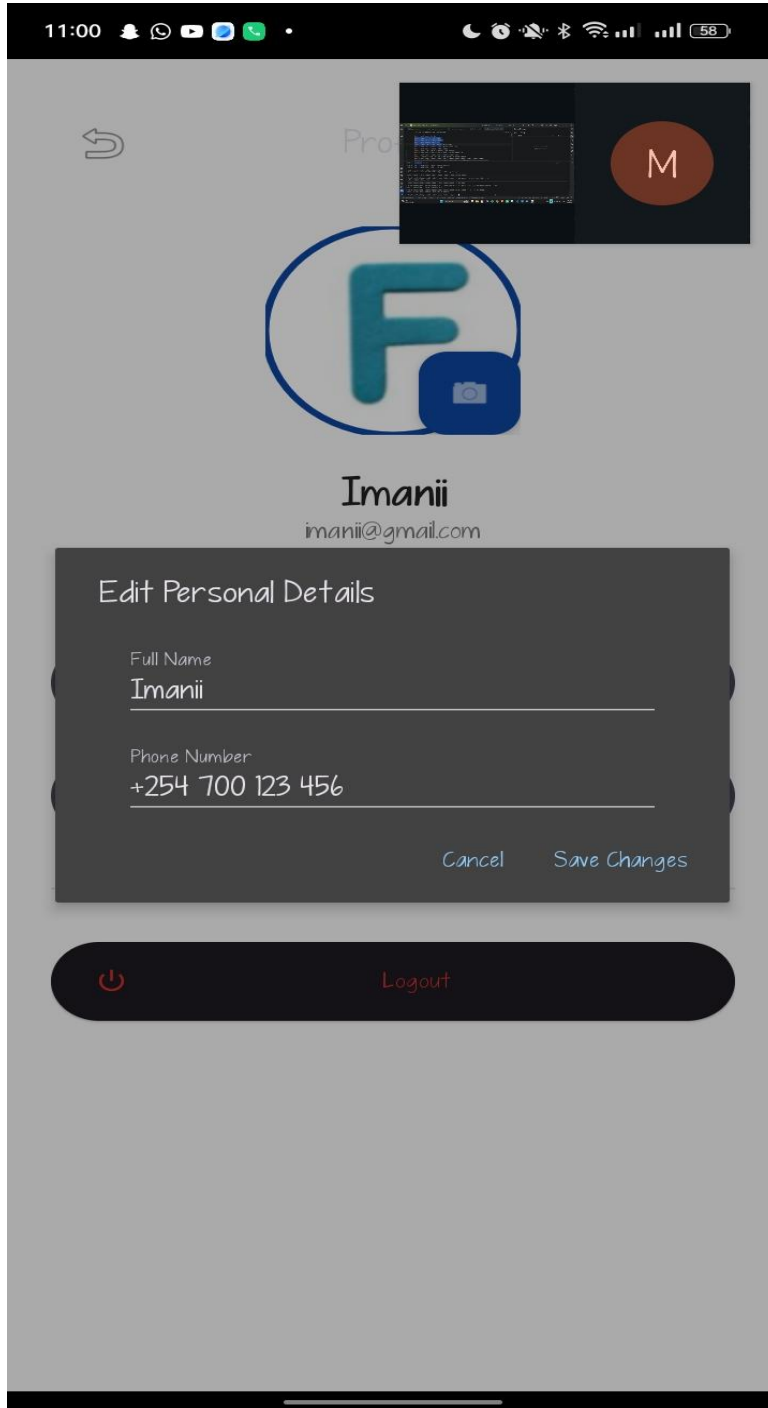
Login

Don't have an account? [Sign Up](#)

Personal reflection

I learned about event handling and user interaction in the fincoop app. I implemented how users interact with the system through buttons, forms, and input fields. This improved my understanding of how applications respond to user actions and enhanced my programming skills.

WEEK 5 DATA MANAGEMENT.



11:01

26:00



Transaction History



Search transactions



Transfer to 07223B169

08 Jun 2026, 18:40

-KES 200



Withdrawal from Savings

08 Jun 2026, 18:39

-KES 200



Deposit to Savings

08 Jun 2026, 18:39

+KES 500



Transfer to 0723402713

08 Jun 2026, 11:30

-KES 500



Withdrawal from Savings

08 Jun 2026, 11:29

-KES 500



Deposit to Savings

08 Jun 2026, 11:29

+KES 1,000



Deposit to Savings

08 Jun 2026, 09:30

+KES 500



Deposit to Savings

08 Jun 2026, 00:23

+KES 1,000



Transfer to +254733817392

08 Jun 2026, 00:17

-KES 1,000



Transfer to 0723402713

08 Jun 2026, 00:17

-KES 11,500

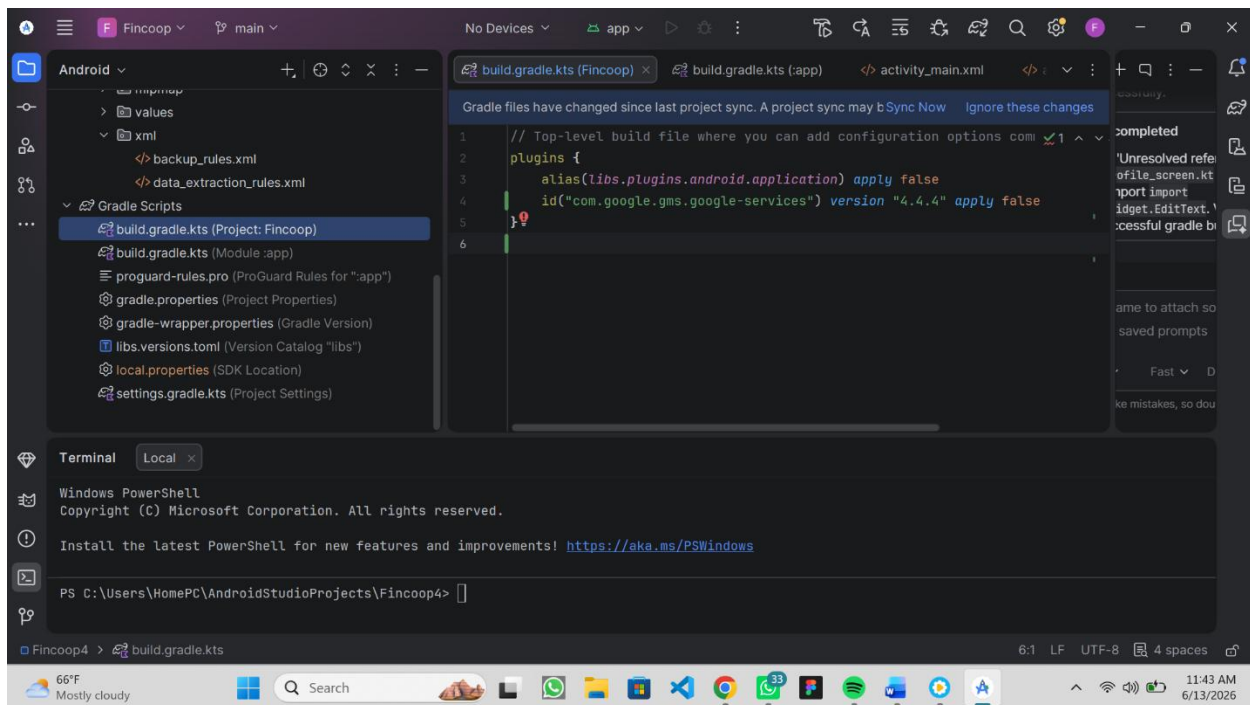


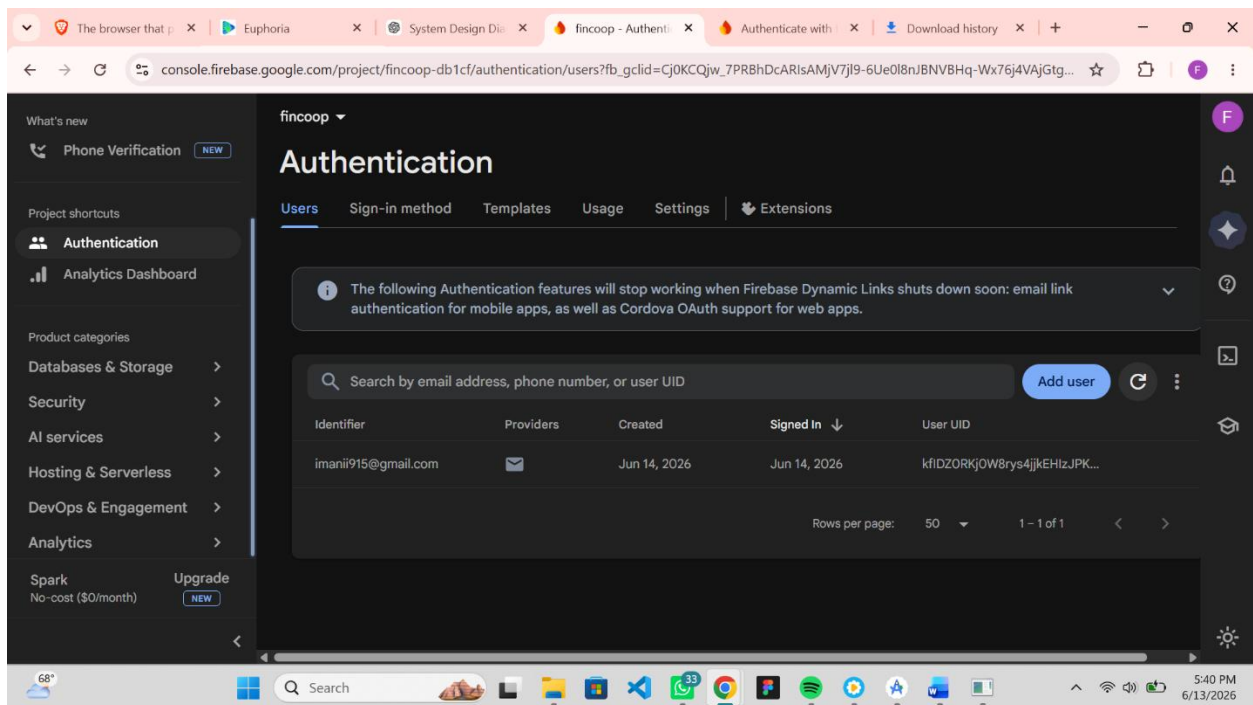
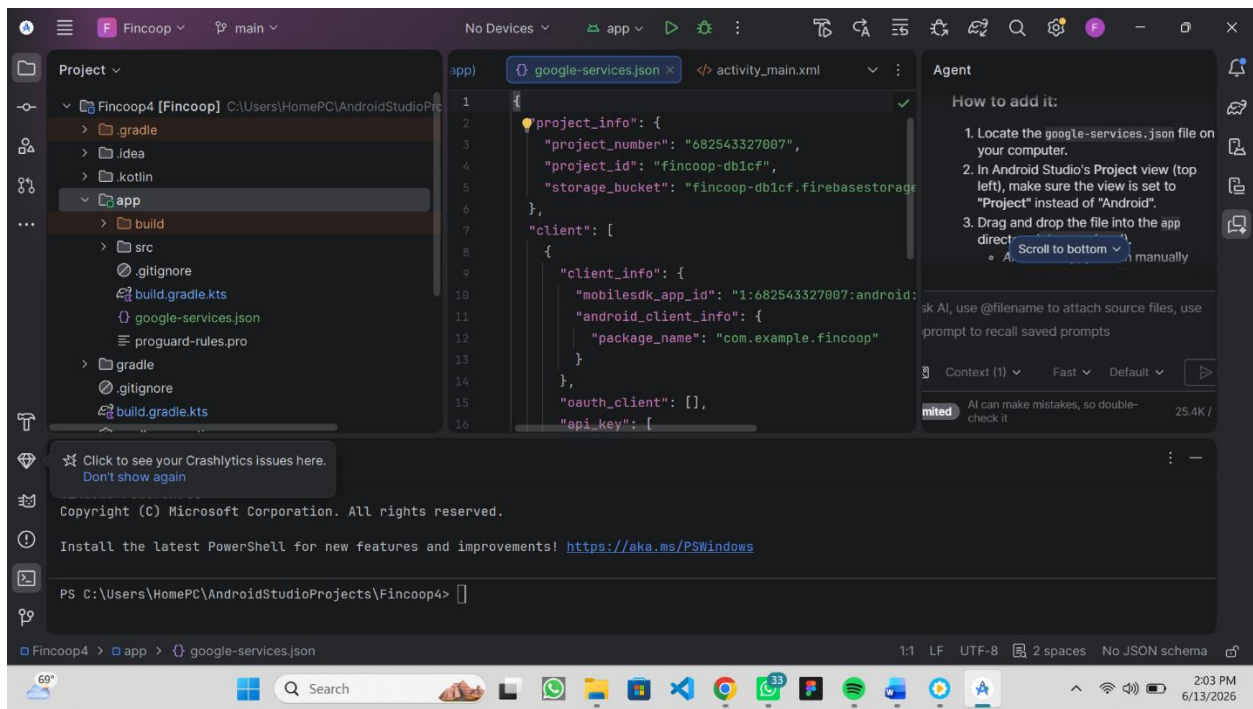
Withdrawal from Savings

Personal Reflection

I focused on data management in the FinCoop application project. Through this role, I learned how to store, retrieve, update, and manage member information, savings records, loan details, and transaction data using local storage and database systems. This experience enhanced my understanding of database management, data security, and efficient record keeping in financial applications. It also improved my technical skills in handling application data and strengthened my ability to develop reliable systems that support accurate and secure financial operations.

WEEK 6:NETWORKING AND API





Firebase console overview for project fincoop. The interface includes a sidebar with navigation links (Project Overview, Settings, What's new, Phone Verification, Product categories) and a main content area. The main content area displays a welcome message "Hello, Faith" and a Gemini chat interface. The Gemini chat interface shows a greeting "Hello, Faith" and a prompt "How can I help you?". A sidebar on the right contains a Gemini chat window with a greeting "Hello, Faith" and a prompt "How can I help you?". The bottom status bar shows the temperature "66°F Mostly cloudy" and the time "11:21 AM 6/13/2026".

Firebase console "Add app" wizard for project fincoop. The wizard is in step 2, "Download and then add config file". It provides instructions for downloading the google-services.json file and adding it to the project root directory. A screenshot of the Android Studio Project view shows the file structure, with the google-services.json file highlighted. The bottom status bar shows the temperature "66°F Mostly cloudy" and the time "11:30 AM 6/13/2026".

Register app
Android package name: com.example.fincoop

2 Download and then add config file
Instructions for Android Studio below | [Unity](#) | [C++](#)

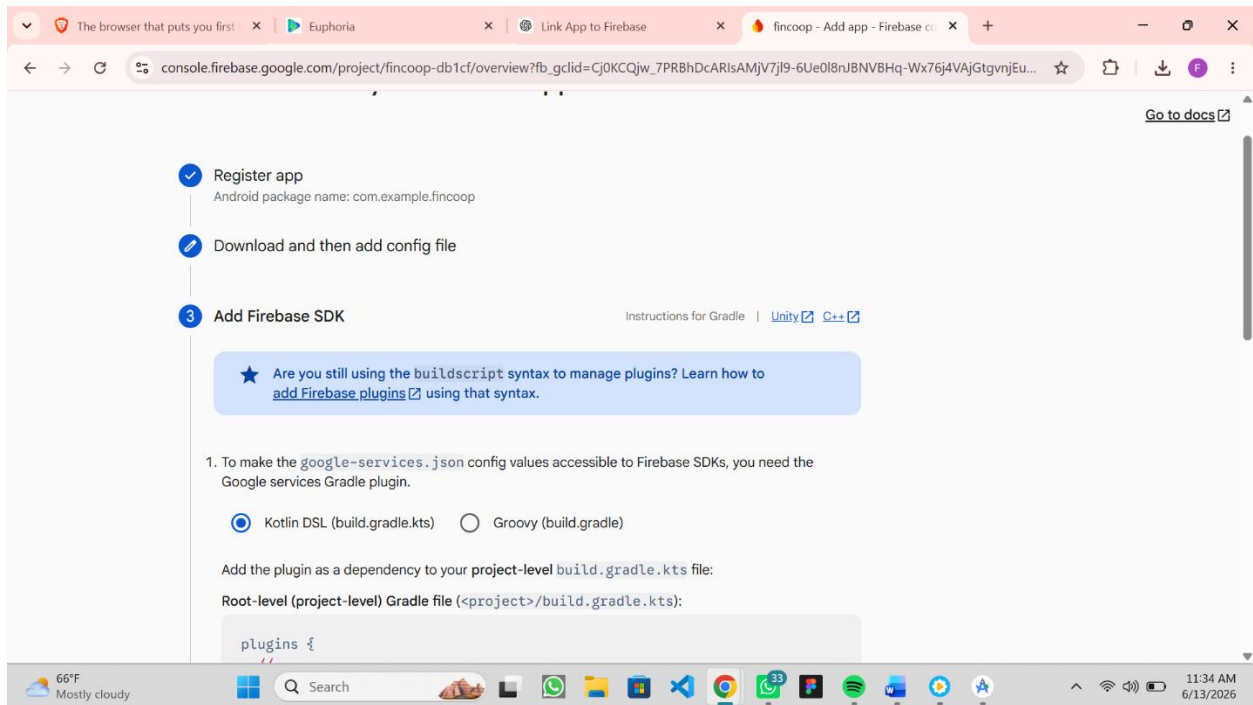
[Download google-services.json](#)

Switch to the Project view in Android Studio to see your project root directory.

Move your downloaded google-services.json file into your module (app-level) root directory.

[Next](#)

3 Add Firebase SDK



My app requires networking because an app needs networking to exchange data with servers, access online services, synchronize information, and enable real-time communication.

Personal Reflection

I focused on implementing networking and API integration for the FinCoop App. During this process, I learned how applications communicate with servers to send and retrieve data through APIs. I gained practical experience in handling network requests, managing data exchange, and ensuring reliable communication between the application and backend services. This enhanced my understanding of networking concepts, API functionality, and system connectivity while improving my problem-solving and software development skills.

